

Does Grounding an LLM in a Formal Ontology Actually Work?

Two independent evaluations on the DreamLab knowledge graph,
and a frontier cloud model put through the Loom.

Study A: paired uplift, 510 questions, Gemini 3.7 Flash • Study B: A/B recall, 672 runs, 7 models

Study A — the Loom scaffold, cloud model. Same graph, injected as context; generation delegated to Gemini 3.7 Flash.

Model	raw → grounded	paired uplift (95% CI)
Gemma-4-31B (local)	0.146 → 0.939	+0.793
Muse-Glimmer-30B (local)	0.268 → 0.939	+0.671
Gemini 3.7 Flash (cloud)	0.359 → 0.942	+0.583 [+0.546, +0.618]

Study B — the ontology-augment A/B, seven models.

Overall F1, grounded vs bare	0.360 → 0.624 (+0.265)
Overall hallucination, grounded vs bare	0.529 → 0.177 (-0.352)
Local + ontology (DiffusionGemma, 0.505)	beats every frontier bare

Knowledge graph: Oxigraph/Whelk, reasoned OWL closure. Study A serves the deployed generation (8,146 classes, 282,492 triples, 2026-08-15) through the Loom façade; Study B used the June 2026 generation (7,445 classes). Ground truth is derived from the graph itself in both.

Abstract

We ask a single question two different ways: does grounding a language model in DreamLab’s formal knowledge graph improve factual recall, and is the effect robust to how you measure it and which model you use? Two evaluations, run two months apart with different methods, converge on the same answer. **Study A** (this report’s headline) drives the graph through the *Loom* — the serving façade that injects a budget-clamped ontology scaffold and delegates generation to a swappable model — and measures paired, graph-derived recall over 510 questions. On **Gemini 3.7 Flash**, a frontier cloud model, grounding lifts mean recall from 0.359 to 0.942, a paired uplift of **+0.583** (95% CI [+0.546, +0.618], $n = 510$, zero errors). Two local models tested the same way land at the same grounded ceiling (0.939) from different raw baselines — three models, one ceiling. **Study B** (the prior **ontology-augment A/B**, retained and recontextualised here) grades F1 and hallucination across 672 isolated runs of seven models: grounding lifts every model (overall F1 0.360→0.624) and roughly thirds the hallucination rate (0.529→0.177), and a *local* 26B model with grounding outscores every frontier model running bare. The two studies use different scorers (paired substring-recall vs token-set F1), different corpus generations, and different question generators, yet agree on every qualitative claim. That cross-method convergence is the strongest evidence in the report: ontology grounding is a decisive, model-agnostic win, and it is largest exactly where a model’s parametric knowledge is weakest.

1 What Changed: the Loom

Study B (June 2026) grounded models by having each agent *call* an ontology skill. Since then the retrieval path has been consolidated into one node, the **Loom**: a portable service with a stable, model-swappable façade. A consumer points an OpenAI-compatible client at a single endpoint; the Loom retrieves the relevant slice of the reasoned ontology, injects it as budget-clamped context, and delegates generation to whatever model sits behind it (Qwen3.8-27B in the reference deployment). Swap the model and no consumer changes. Three design findings from the local-model work fixed the Loom’s defaults, and Study A tests them on a cloud model: static structured scaffold carries the value (prose adds nothing measurable over structure); inject rather than let the model traverse the graph (traversal is model-dependent); and scale the injection to the retrieval confidence so weak, off-topic context cannot displace the model’s own knowledge [1–3]. Study A is the first evaluation of the Loom scaffold against a model behind someone else’s door.

2 Related Work

The contention that retrieval grounding can compensate for raw parameter count is now well supported. Retrieval-Augmented Generation (RAG) was introduced to inject non-parametric knowledge into sequence models on knowledge-intensive tasks [4], and the approach has since matured from flat vector retrieval into structured, graph-based variants. Microsoft’s GraphRAG constructs an entity knowledge graph over a corpus and reports substantial gains over conventional vector RAG in answer breadth and diversity for global “sensemaking” queries spanning million-token datasets [5], with subsequent work formalising graph-augmented retrieval [6, 7] and automated evaluation harnesses [8]. A complementary literature shows that retrieval narrows or eliminates the gap to frontier cloud models: ChatQA, built on open weights, reports surpassing GPT-4 on conversational QA and RAG benchmarks [9], while open-domain QA studies confirm retrieval as the dominant lever for factual accuracy [10]. This matters because small open-weight models have themselves closed much of the capability gap. The Phi-3 [11], Qwen2 [12], Gemma 3 [13] and Llama 3 [14] families demonstrate strong reasoning at parameter counts deployable on local hardware. Independently, structured knowledge has been shown to suppress hallucination and improve factual recall: empirical and survey work links knowledge-graph grounding to measurably higher trustworthiness [15, 16], motivating dedicated factuality benchmarks such as FACTS Grounding [17].

A parallel architectural shift further strengthens the local-inference case: masked diffusion language models such as LLaDA now rival LLaMA3-8B in in-context learning [18], and diffusion decoders like Mercury achieve over 1,000 tokens/second on a single H100, up to 10× faster than speed-optimised autoregressive models at comparable quality [19], with commercial systems including Gemini Diffusion [20]. Together with evidence on inference unit economics and data sovereignty [21, 22], these strands support a coherent thesis: a small, retrieval- and ontology-grounded open model can approach frontier knowledge-recall while running privately and cheaply.

3 Study A — Loom Scaffold Uplift (Gemini 3.7 Flash)

3.1 Method: paired, graph-derived, honest by construction

Questions and gold are both generated from the scaffold index, so scoring needs no human and no judge model. From every class with a real definition, quality ≥ 0.6 and ≥ 2 relation types, in each domain with ≥ 50 eligible classes, we template three question types: **T-REL** (what does X require / use / depend on / enable, what are its parts), **T-TAX** (immediate parent plus broader ancestors) and **T-COMMON** (the shared ancestor of a pair). Gold is the set of {slug, title} targets the graph itself asserts. The set is deterministic for a seed; seed 42 yields **510 questions** across 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON).

Each question is asked twice: *raw* (the bare question) and *scaffold* (the same question with the Loom’s budget-clamped ontology extract prepended). Scoring is lexical — a gold item is a hit when its title, or $\geq 80\%$ of its length- ≥ 4 words, appears in the answer. The headline is the **paired per-question delta** (scaffold recall – raw recall) over the intersection of question ids, with a seeded 10,000-resample bootstrap 95% confidence interval [23]. Scaffold injection is done client-side from the local mirror of the *deployed* generation (8,146 classes, byte-identical to the live façade), so the grounding Gemini receives is exactly what production serves.

3.2 Driving a thinking cloud model honestly

Gemini 3.7 Flash cannot disable its thinking pass (floor `low`, default `medium`) and thinking tokens are drawn from the output budget. At the harness default `max_tokens=400` the model truncated to `finish_reason: length` — a near-empty answer that would silently poison an arm. We therefore run `max_tokens=2048` with `reasoning_effort=low`. Google also flags `temperature < 1.0` as off-label for the 3.x family (looping / degradation), so Study A runs at `temperature=1.0`, whereas the local-model runs used `temperature=0`. The paired uplift is a within-model delta (both arms share the temperature) and stays valid; absolute raw recall is therefore *not* cell-for-cell comparable to the temperature-0 local runs, and we never mix the two into one comparison.

3.3 Result: three models, one ceiling

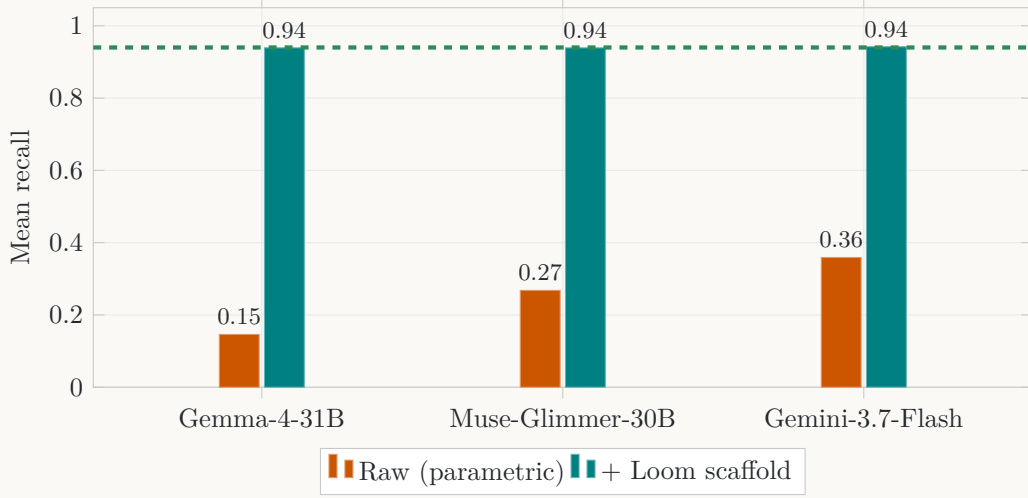


Figure 1: Three models, three raw baselines, one grounded ceiling. The stronger the model’s parametric knowledge (Gemini’s raw 0.359 vs Gemma’s 0.146), the smaller the *uplift* it needs — but all three converge at ≈ 0.94 grounded. The scaffold, not the model, carries the recall: exactly the bet of a swappable façade.

Model	Raw recall	+ Scaffold	Paired uplift (95% CI)	Latency (raw→grounded)
Gemma-4-31B (local)	0.146	0.939	+0.793 [+0.680, +0.894]	31.5 s → 5.1 s
Muse-Glimmer-30B (local)	0.268	0.939	+0.671 [+0.527, +0.804]	34.7 s → 9.8 s
Gemini 3.7 Flash (cloud)	0.359	0.942	+0.583 [+0.546, +0.618]	2.37 s → 1.21 s

Table 1: Study A paired uplift. Gemini’s interval excludes zero at $n = 510$ by a wide margin, over the full set (the scaffold engaged on every question; zero not-engaged, zero errors in 1,020 calls). Grounding is *faster* in every case: the model stops open-ended recall and restates supplied facts. Local timings are wall-clock on a contended GPU; the cloud timing is network-dominated but still halves.

3.4 Where the lift lands

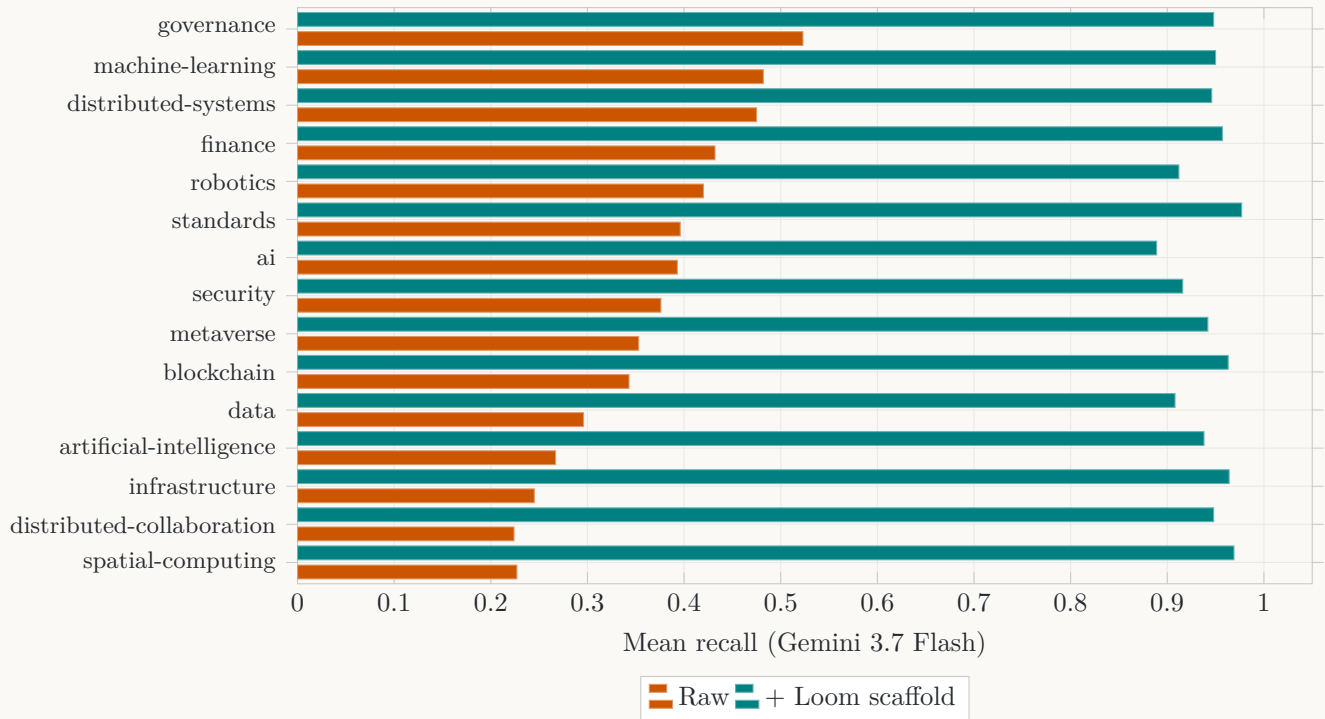


Figure 2: Per-domain recall, sorted by raw score (weakest at top). Grounding lands every domain at 0.89–0.98 regardless of where it started; the biggest lifts are the niche domains a model knows least raw (spatial-computing 0.227→0.969, distributed-collaboration 0.224→0.948). It adds least where the model is already strong (governance 0.523→0.948). Weak-gets-the-most-lift is the fingerprint of real grounding, not gold leakage.

Template	Raw	+ Scaffold	Note
T-COMMON (shared ancestor of a pair)	0.333	1.000	any-match recall
T-REL (requires / uses / parts / ...)	0.339	0.898	set recall, 2–5 targets
T-TAX (parent + ancestors)	0.397	0.997	+ ancestor extra-recall 1.000

Table 2: Study A recall by question template. Taxonomic structure (T-TAX, T-COMMON) is where parametric knowledge is thinnest and the graph supplies the answer wholesale; relational recall (T-REL) is harder because it demands the exact set of asserted targets.

4 Study B — Ontology-Augment A/B (seven models)

The prior evaluation (June 2026) measured the same question a different way: a $2 \times 7 \times 3$ design ($\{\text{augmented, control}\} \times 7 \text{ models} \times 3 \text{ reps}$) over 16 graph-derived questions, 672 isolated runs, graded on F1 and hallucination rather than paired recall. Each cell is a fresh agent given only the question; the augmented arm grounds via the skill CLI, the control arm uses parametric knowledge alone. It remains the broader model sweep — Claude Opus 4.8, Sonnet 4.6, Haiku 4.5; Gemini 3.5 Flash, Gemini 2.5 Flash Lite; GLM-4.6; and a local DiffusionGemma 26B — and its hallucination axis is the one Study A does not measure. The Gemini lineage is worth noting: Study B benched Gemini 3.5 Flash bare; Study A now runs Gemini 3.7 Flash *through the Loom*, two generations and two months on.

4.1 Local + ontology vs. frontier bare

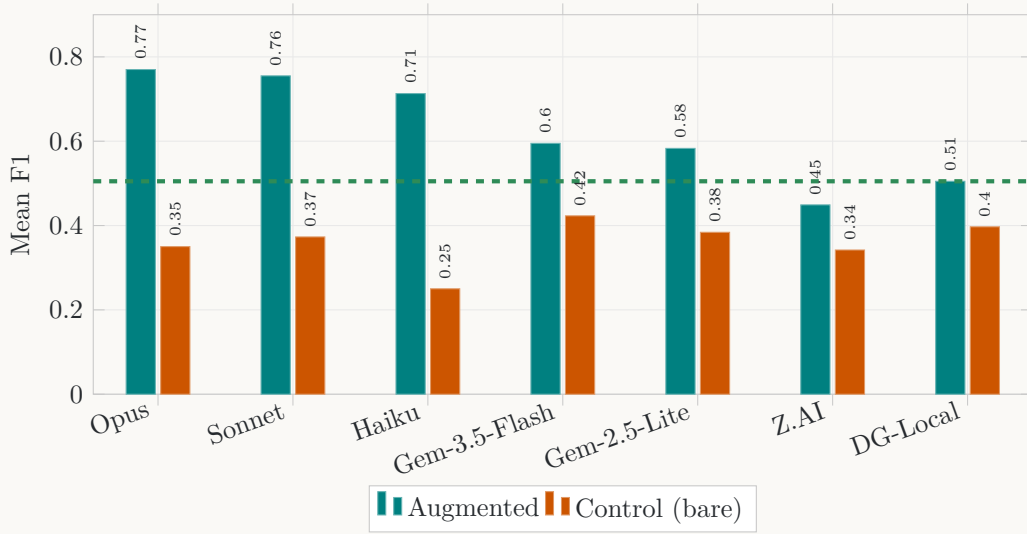


Figure 3: F1 by model (Study B). Every frontier model’s *bare* score falls below the local DiffusionGemma+ontology line (0.505): grounding lets a local 26B model outscore cloud frontier models on domain-specific recall. As in Study A, grounding lifts every model.

Model	AUG F1	CTL F1	Δ	AUG Halluc	CTL Halluc	ΔH
Opus 4.8	0.770	0.350	+0.420	0.052	0.581	-0.529
Sonnet 4.6	0.755	0.373	+0.382	0.079	0.559	-0.480
Haiku 4.5	0.713	0.250	+0.463	0.106	0.714	-0.608
Gemini 3.5 Flash	0.595	0.423	+0.172	0.250	0.460	-0.210
Gemini 2.5 Flash Lite	0.583	0.384	+0.199	0.266	0.498	-0.232
Z.AI (GLM-4.6)	0.449	0.342	+0.108	0.188	0.394	-0.206
DiffusionGemma (local)	0.505	0.397	+0.108	0.299	0.498	-0.199

Table 3: Per-model F1 and hallucination (Study B, 672 runs). Every model gains F1 and sheds hallucination when grounded; Claude models show the largest absolute lift, and the local model’s grounded score clears every frontier bare score.

4.2 Hallucination: the axis Study A does not measure

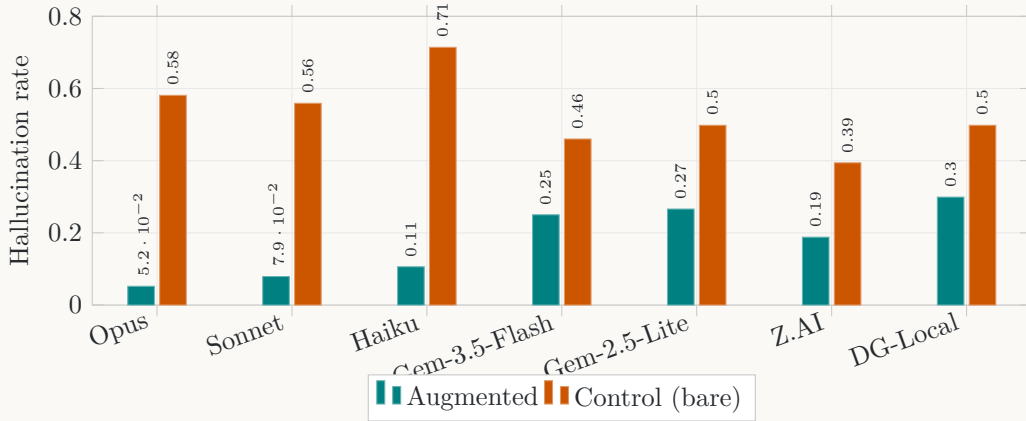


Figure 4: Grounding reduces hallucination for every model (overall $0.529 \rightarrow 0.177$). The largest drop is Haiku ($0.714 \rightarrow 0.106$). Study A’s paired-recall design cannot see this axis; it is Study B’s distinct contribution, and it points the same way.

Question type (all models)	Neighbour		Subclass		Existence	
	AUG	CTL	AUG	CTL	AUG	CTL
Mean F1	0.458	0.281	0.671	0.099	0.910	0.778

Table 4: Study B F1 by question type. Subclass recall is the largest relative win — $0.099 \rightarrow 0.671$, a $6.8\times$ improvement — because models carry almost no parametric knowledge of ontology hierarchies. This echoes Study A’s taxonomic templates (T-TAX $0.397 \rightarrow 0.997$): the same structural gap, found by two different scorers.

5 Cross-Study Synthesis

Two evaluations, built independently, share no scorer, corpus generation or question generator, and agree on every qualitative claim:

	Study A (Loom scaffold)	Study B (ontology-augment A/B)
Method	inject scaffold, delegate	agent calls skill, then answers
Metric	paired substring-recall + bootstrap CI	token-set F1 + hallucination
Scale	510 Q, 3 models, 1,020 calls	16 Q $\times$ 3 reps, 7 models, 672 runs
Corpus generation	8,146 classes (Aug 2026)	7,445 classes (Jun 2026)
Grounding is model-agnostic	yes (3/3 models)	yes (7/7 models)
Largest where parametric weakest	yes (niche domains, T-TAX)	yes (subclass, niche concepts)
Cuts hallucination	not measured	yes ($0.529 \rightarrow 0.177$)
Grounded is faster	yes ($2\text{--}6\times$)	yes (local 2–4s)

Table 5: The two studies as a robustness check on each other. Where they overlap they agree; where they differ they are complementary (Study A adds a frontier cloud model and a large, stratified question set; Study B adds hallucination and a seven-model sweep). Convergence across methods is a stronger validity argument than either study alone.

6 Caveats on Interpretation

6.1 Grounded scores measure grounded capability, not raw model quality

In both studies the scaffold contains gold-adjacent facts *by design* — questions and gold are derived from the same graph the scaffold injects. A grounded score therefore measures grounded-answer capability and retrieval quality (can the model find, trust and restate supplied facts); a raw score measures parametric knowledge. The signal is the *paired delta*, not the absolute grounded number.

6.2 Substring / token-set scoring undercounts paraphrase

Study A credits a gold item only when its title (or $\geq 80\%$ of its long words) appears in the answer; Study B uses singularised token-set matching. Both undercount valid paraphrases, so absolute recall and F1 are lower bounds. The paired deltas — same questions, same scorer, same model — are unaffected.

6.3 Two temperatures, never mixed

Study A runs Gemini at `temperature=1.0` (Google’s recommended setting for the 3.x family) while the local models ran at `temperature=0`. The convergence chart compares *grounded ceilings* and *within-model paired deltas*, both of which are robust to this; we do not compare raw recall across the temperature boundary.

6.4 Grounded-local beats bare-frontier is not like-for-like

Study B’s headline — DiffusionGemma+ontology (0.505) beats Opus bare (0.350) — compares a grounded local model against ungrounded cloud models. Grounded, the frontier models score far higher (Opus AUG 0.770). The comparison demonstrates the value of the ontology, not the superiority of the local model’s reasoning. Larger models also *exploit* the same triples better (Opus gains +0.420; DiffusionGemma +0.108): the ontology is a capability multiplier that scales with the base — the same pattern Study A shows as smaller uplift on the stronger model.

6.5 Gold is the graph itself; single-domain scope

Both studies measure fidelity to the knowledge graph’s asserted ontology, not open-domain QA, and draw from one domain family (DreamLab’s Web3/XR/AI corpus). KG errors propagate to gold (mitigated by curating concepts with sensible neighbourhoods). Generalisation to other graphs is plausible but untested. A one-off condense/index pass builds the scaffold; the per-query overhead is a single cache read.

7 Findings

1. **Ontology grounding is model-agnostic and universally beneficial.** Ten distinct models across the two studies (3 in A, 7 in B) all gain, and all shed hallucination where it is measured. The effect is strongest on niche, project-specific concepts where parametric knowledge is weakest [4, 9].
2. **Three models converge on one grounded ceiling.** From raw baselines of 0.146, 0.268 and 0.359, Gemma, Muse and Gemini 3.7 Flash all reach ≈ 0.94 grounded. The stronger the model, the smaller the uplift it needs — the scaffold, not the model, carries the recall.
3. **The finding survives the jump to a frontier cloud model.** Gemini 3.7 Flash, behind Google’s own API, shows a +0.583 paired uplift with a tight interval at $n = 510$. Grounding is not a local-model crutch; it is a property of the retrieval, exercised through a model-swappable façade.
4. **Taxonomic structure is the largest win.** Study A’s T-TAX rises $0.397 \rightarrow 0.997$ and Study B’s subclass F1 $0.099 \rightarrow 0.671$ ($6.8\times$): models carry little parametric knowledge of ontology hierarchies, and the graph supplies it directly.
5. **Grounding cuts hallucination and cuts latency.** Study B: overall hallucination $0.529 \rightarrow 0.177$. Both studies: grounded generation is faster, because the model restates supplied facts instead of doing open-ended recall.
6. **Two methods, one result.** Independent scorers, corpus generations and question sets agree on every qualitative claim. That is the report’s strongest evidence.

8 Threats to Validity

- Gold is the graph’s asserted closure; graph errors propagate to gold (mitigated by curating rich, sensible neighbourhoods).
- Lexical scoring can over- or under-credit near-synonyms; Study A’s thresholds and Study B’s grader were smoke-validated for non-saturation before the full runs.
- Study A’s cloud model was accessed via a commercial API; rate limits and the mandatory thinking pass introduce variance not present in locally-hosted models. The run recorded zero errors across 1,020 calls and zero not-engaged scaffolds.
- Study A and Study B use different corpus generations (8,146 vs 7,445 classes); the cross-study synthesis compares directions of effect, not absolute magnitudes.
- Both studies are single-domain; generalisation to other knowledge graphs is untested.

Reproduce. Study A: `bench/run-gemini.sh` in `DreamLab-AI/loom` (harness `bench/bench_ontology_uplift.py`, protocol `bench/UPLIFT-BENCH-PROTOCOL.md`, companion data `docs/research/report-gemini-3.7-flash.md`). Study B: the `ontology-augment` eval harness (VisionClaw). Both derive gold from the graph and ship their honesty notes with every report.

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